Hadoop & Spark - ISEN 2025/2026

Chapter 1: What is 'Big Data' ?

Goals of this chapter:

In this chapter we will see:

- Different technologies
- Big Data definition

Technologies

Let's start with the technologies!

Question time: Could you give me a list of technologies or tools useful for Big Data?

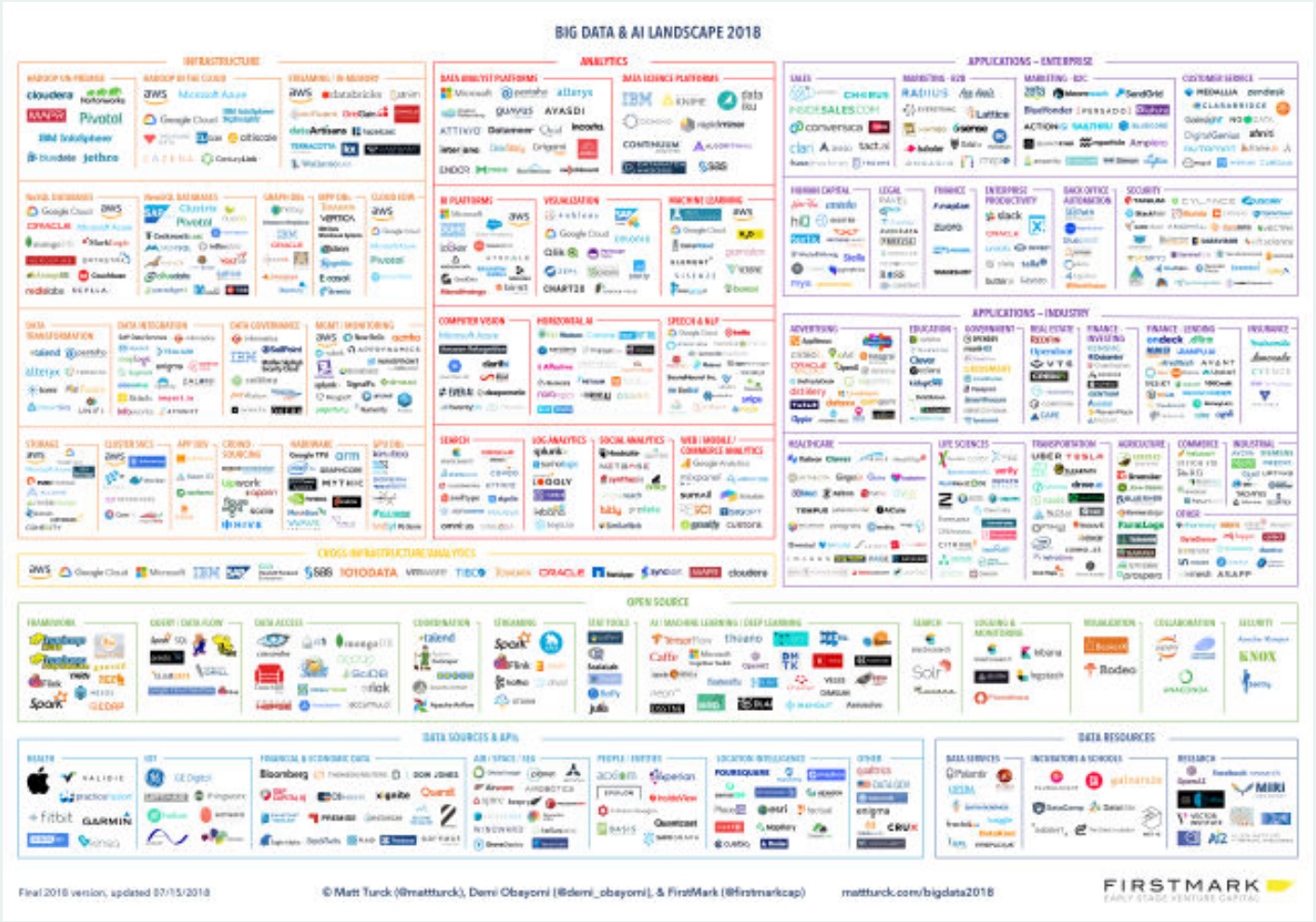


Technologies

Matt Turck and its company Firstmark publish the state of Data and AI every year as a graphic.

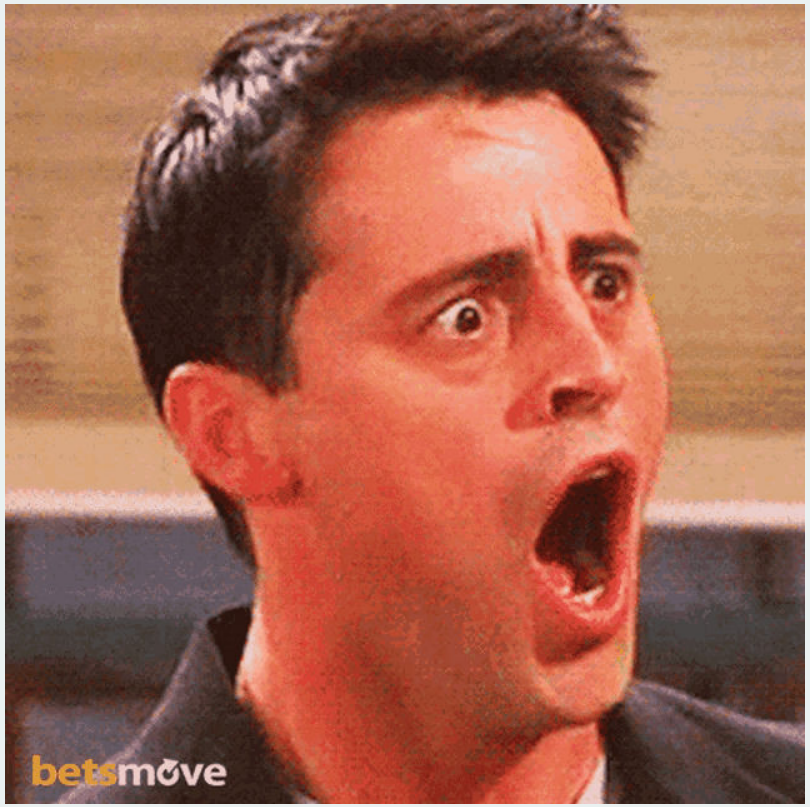


It is big!

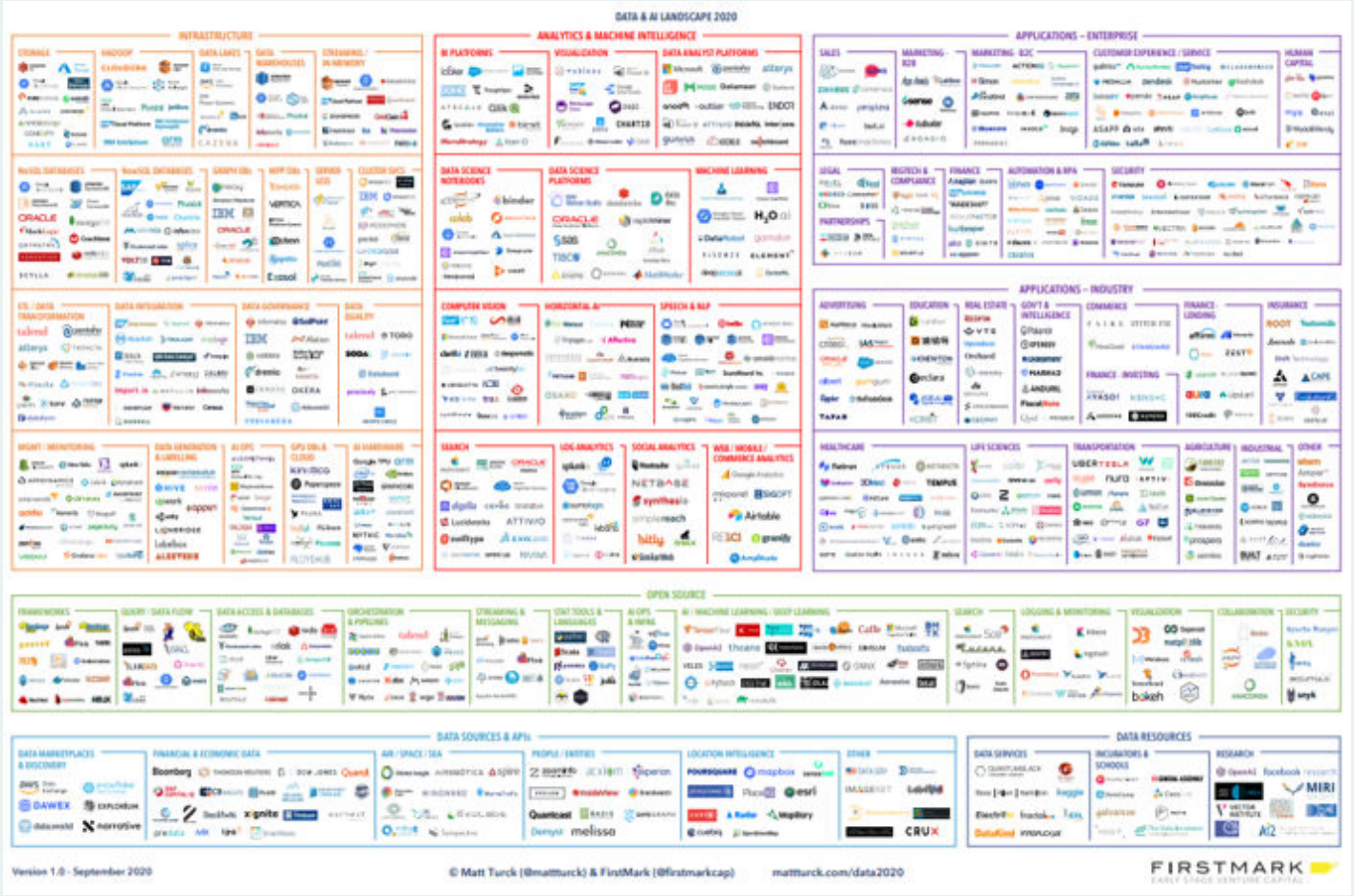

BIG DATA & AI LANDSCAPE 2018
INFRASTRUCTURE
ANALYTICS
APPLICATIONS – ENTERPRISE
APPLICATIONS – INDUSTRY
OPEN SOURCE
DATA SOURCES & APIs
DATA RESOURCES
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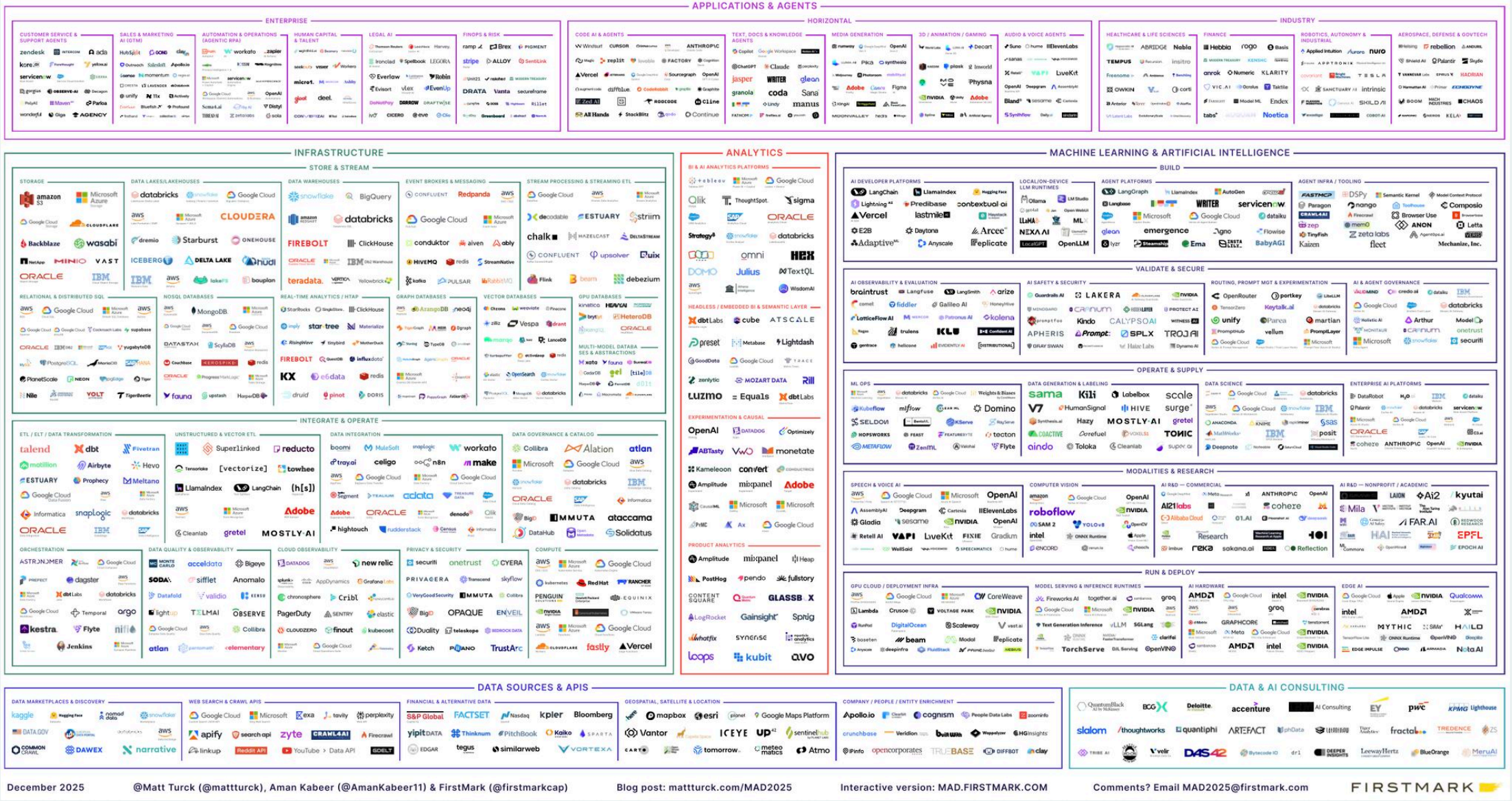
But this is outdated ...

This was in 2018



2020


DATA & AI LANDSCAPE 2020
INFRASTRUCTURE
ANALYTICS & MACHINE INTELLIGENCE
APPLICATIONS – ENTERPRISE
APPLICATIONS – INDUSTRY
OPEN SOURCE
DATA SOURCES & APIs
DATA RESOURCES
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APPLICATIONS & AGENTS
ENTERPRISE
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December 2025
@Matt Turck (@mattturck), Aman Kabeer (@AmanKabeer11) & FirstMark (@firstmarkcap)
Blog post: mattturck.com/MAD2025
Interactive version: MAD.FIRSTMARK.COM
Comments? Email MAD2025@firstmark.com
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Technology grows fast, very fast

The point is:

In this course, concepts are often more important than technologies as they are evolving quickly.

Technology will not make you go far if you do not understand the concepts as a whole.

(But that's no reason not to know how to use it or code properly!)

2025 article -> here



Big Data: Why do we need it?

A few examples



Big Data example 1: Search engine



Current storage = 15 exabytes

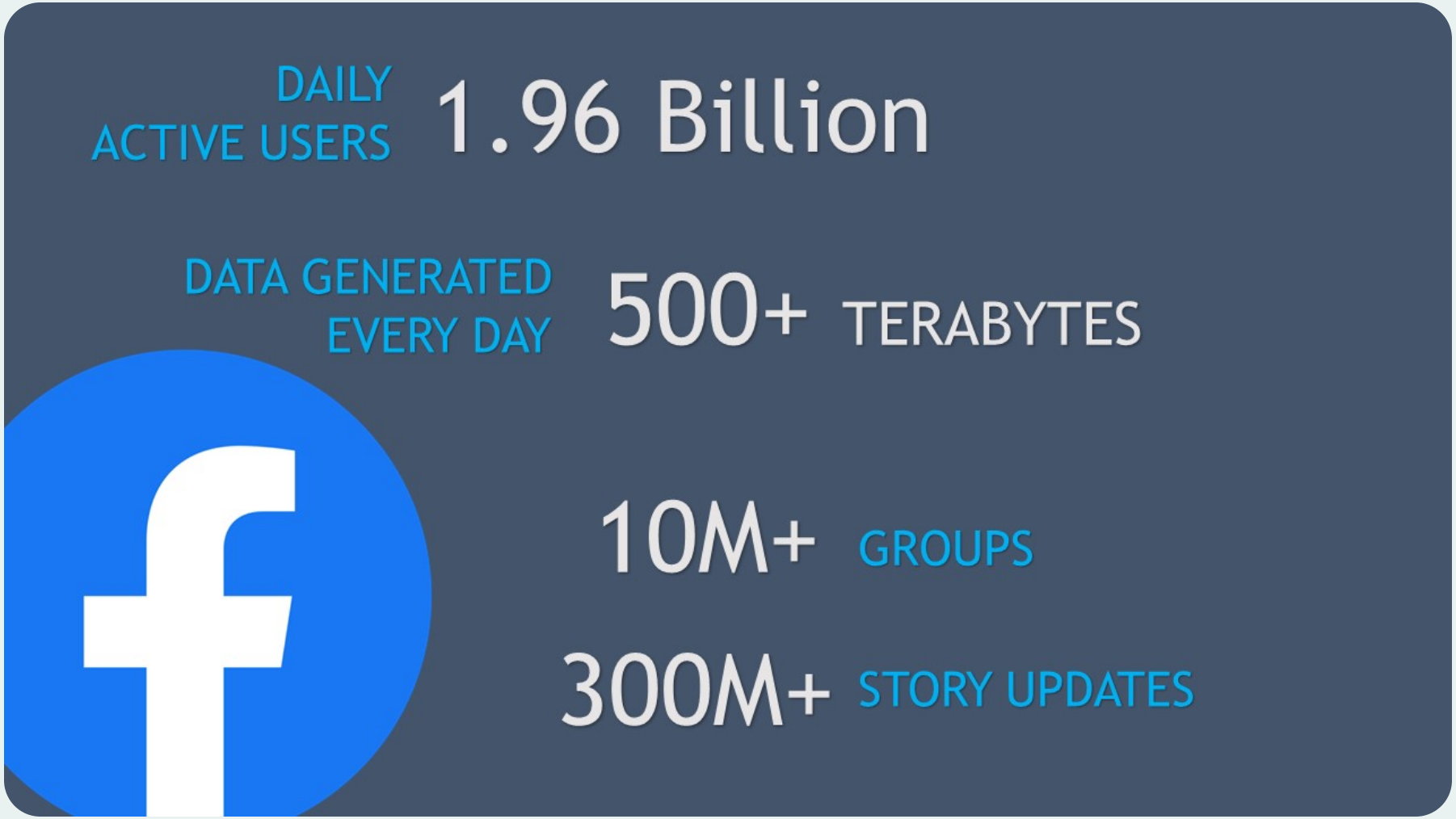
Processed per day = 100 petabytes

Number of pages indexed = 60 trillion

Unique search users per month > 1 billion

Searches per second = 2.3 million

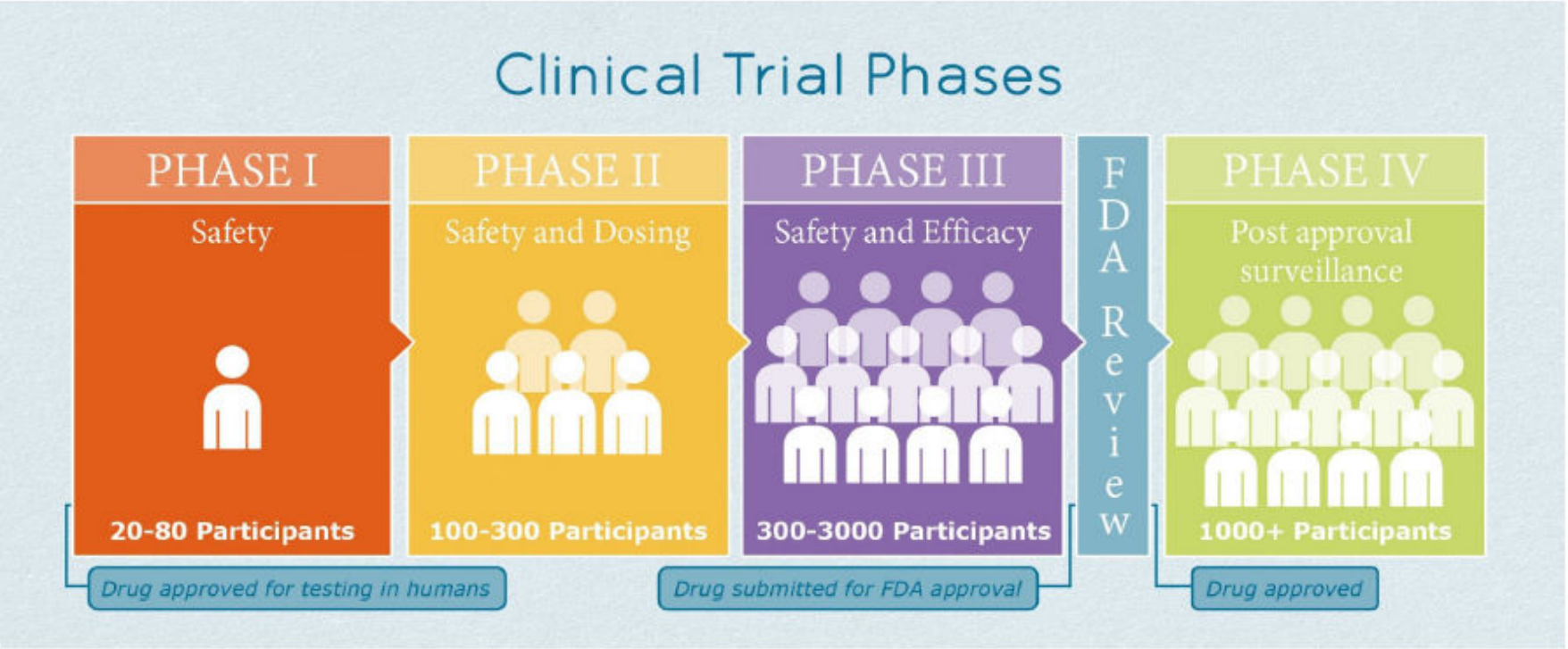
Big Data example 2: Social Network



Big Data example 3: Pharma

Pharmacovigilence DB
Millions of records worldwide

Drug design
Smart Alerting system
Better driving R&D
Personal medicine



Big Data example 4: Surveillance (sadly !)



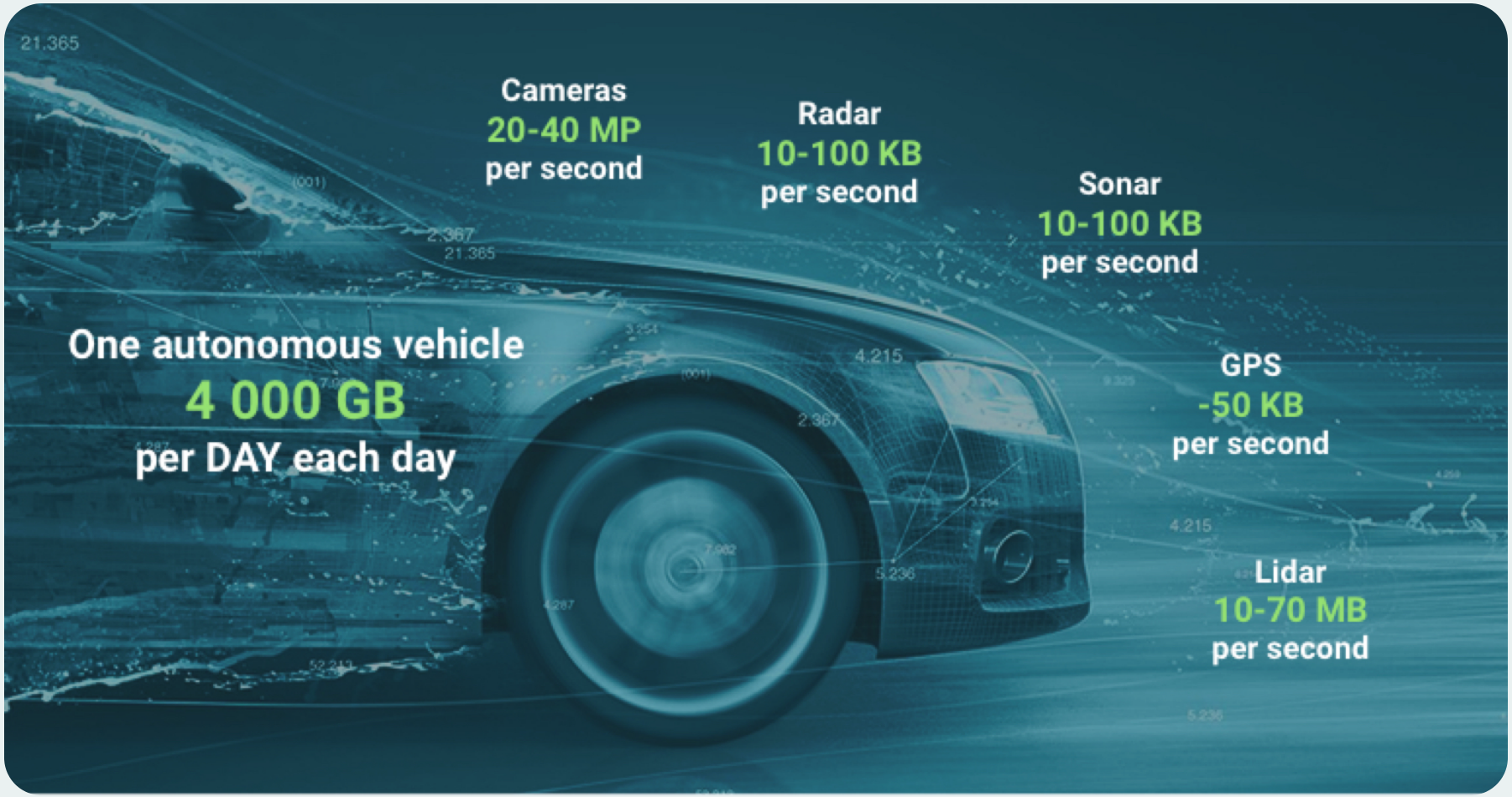
Current storage = ~5 exabytes

Processed per day = 30 petabytes

NSA touches 1.6% of internet traffic per day

Web searches, websites visited, phone calls, credit/debit card transactions, financial and health information

Big Data example 5: Autonomous cars



Big Data and AI is the center of many buzz and trends thanks to AI

One article from Forbes: here

> AI market size is expected to reach $407 billion by 2027

> AI will have an estimated 21% net increase in the United States GDP by 2030

> One in 10 cars will be self-driving by 2030

> 400 million workers could be displaced because of AI

And so on ...

Big Data definition

How do you define it?



Big Data definition

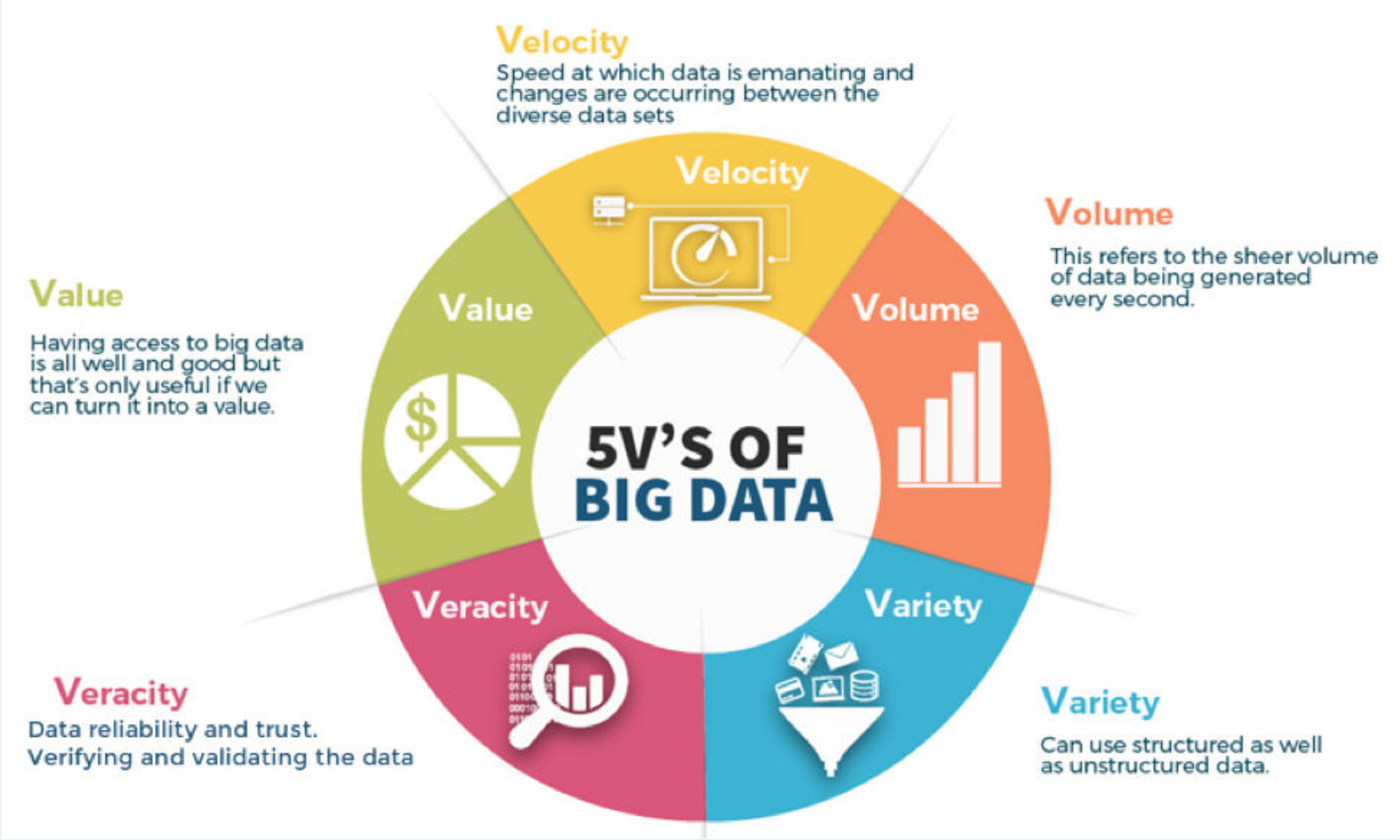
You often see this 3V definition in your courses (Or at least I did!)



Big Data definition

I think this is a better representation.

❗ **Those 5V's are very important to keep in mind** ❗



Credit and used resources:

- Hadoop & Spark courses from previous years by Olivier Moïse (2022) & Fabrice Woittequand (2020)"